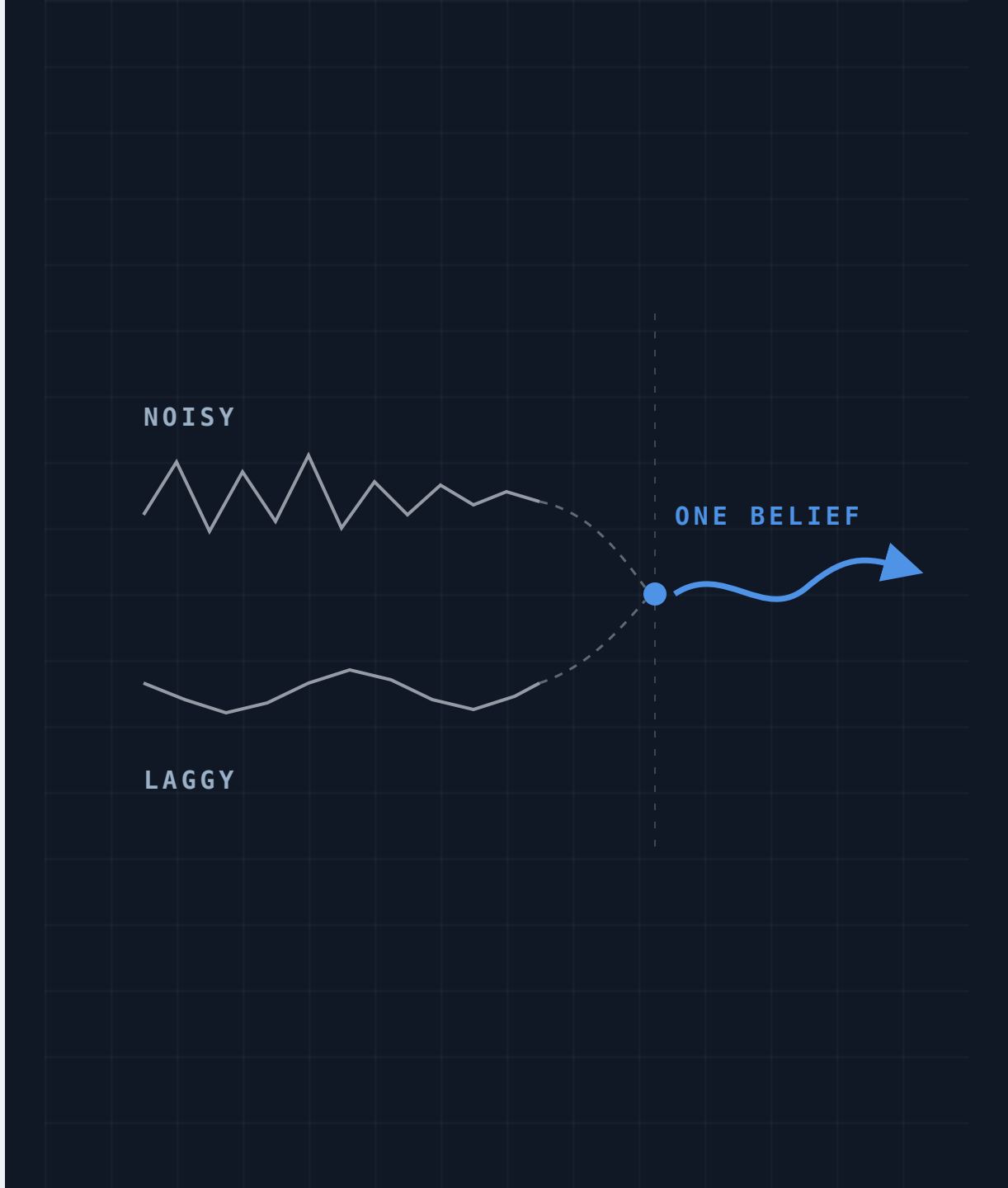
CS 460/560 · AUTONOMOUS ROBOTICS · LECTURE 07

A robot never perceives the world. It reconciles competing estimates.

Every number a robot acts on is a belief: a prediction it already held, nudged by a measurement it does not fully trust.



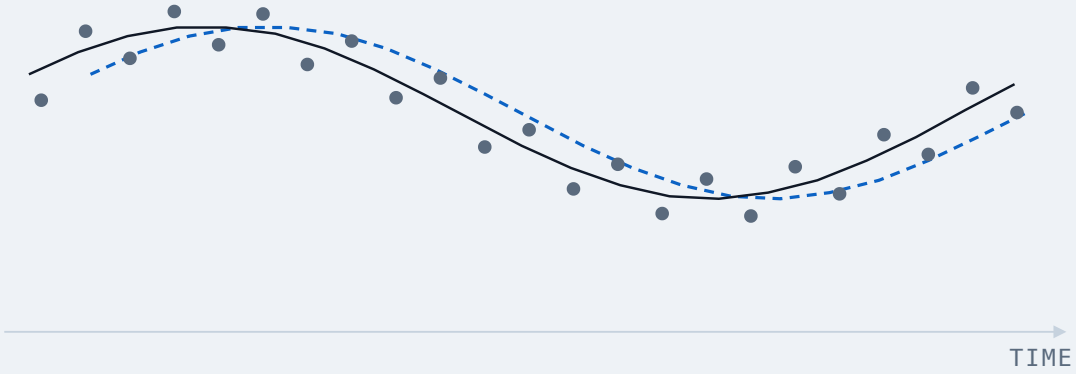
SENSOR FUSION · PREDICT · CORRECT · THE GAIN K

02 · THE PROBLEM

Two estimates. Neither one is the truth.

One sensor answers instantly and **jitters**. The other reports the shape of the motion faithfully but arrives **late**.

Averaging them does not give you the truth. It gives you a third wrong answer, with no record of how wrong.

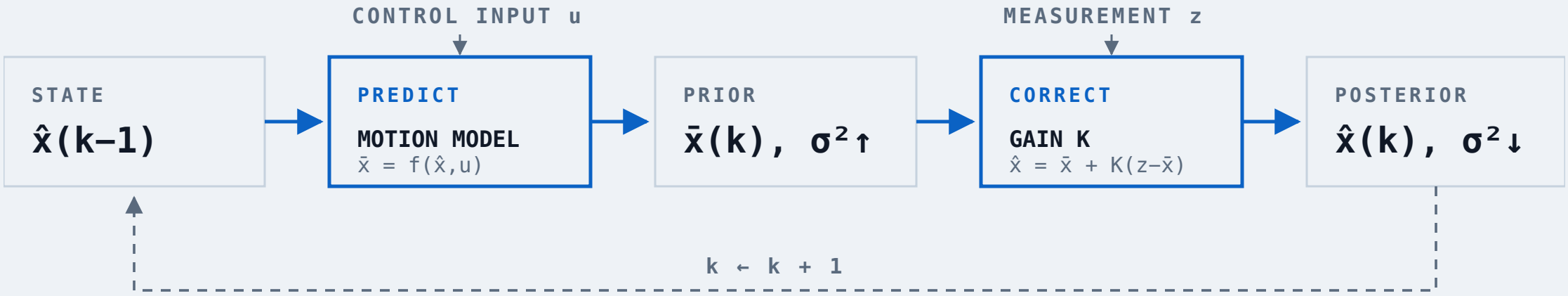


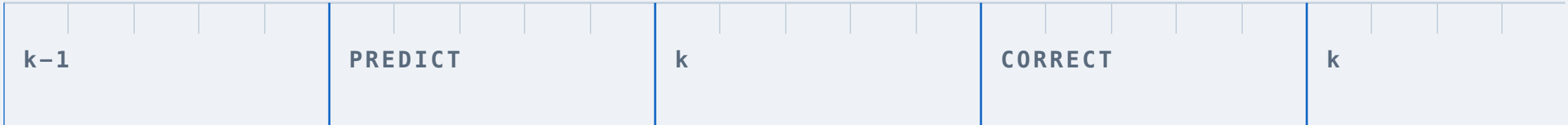
— TRUTH (NEVER OBSERVED) · ● NOISY SENSOR · – – DELAYED SENSOR

03 · THE WHOLE IDEA, ONE LINE

Keep a belief. Predict it forward. Correct it with evidence.

Two steps, repeated at every tick of the clock. Nothing else in the filter is new.





04 · THE GAIN IS THE WHOLE CONTROLLER

One number decides how much of the sensor you swallow.

$$\hat{x} = \bar{x} + K(z - \bar{x})$$

RESIDUAL (Z − X̄) — HOW WRONG THE PREDICTION TURNED OUT TO BE

$\bar{x}$	PREDICTED STATE, BEFORE THE SENSOR SPEAKS
z	WHAT THE SENSOR ACTUALLY REPORTED
K	GAIN, 0 → 1 : THE FRACTION OF THE GAP YOU ACCEPT

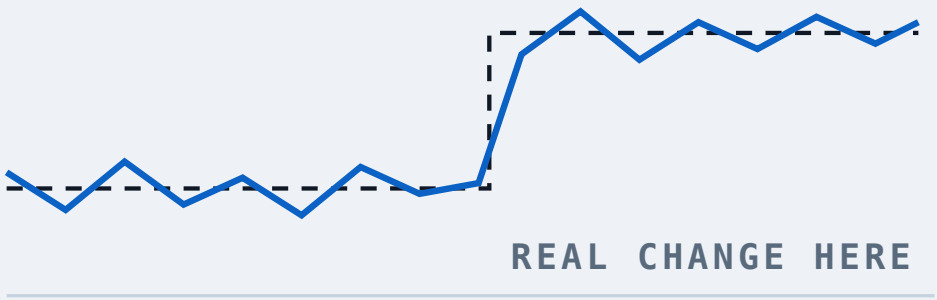
The residual is the only new information in the update. Everything the filter does is decide what fraction of it to believe.

Set **K = 0** and the sensor is ignored. Set **K = 1** and the prediction is thrown away. Every useful robot lives strictly in between.

K buys responsiveness and pays for it in noise.

K ≈ 1

TRUST THE SENSOR



Follows every real change the instant it happens — and inherits every bit of the sensor's jitter. Control built on it chatters.

K ≈ 0

TRUST THE MODEL



Beautifully smooth — and blind to anything the model did not predict. A disturbance is noticed only after the damage.

06 · WHERE K COMES FROM

K is not a taste setting. It falls out of the two variances.

$$K = \sigma^2_{pred} / (\sigma^2_{pred} + \sigma^2_{meas})$$

SENSOR GETS NOISIER

$\sigma^2_{meas} \uparrow \implies K \to 0$

The reading is barely evidence, so keep the belief you already had.

SENSOR GETS CLEANER

$\sigma^2_{meas} \downarrow \implies K \to 1$

Snap to the measurement; your model has nothing better to offer.

MODEL GETS SHAKIER

$\sigma^2_{pred} \uparrow \implies K \to 1$

You have lost the state, so let the sensor pull you back onto it.

MODEL GETS SOLIDER

$\sigma^2_{pred} \downarrow \implies K \to 0$

Hold the belief; one surprising reading should not move you far.



07 · THE TWO STEPS, HONESTLY ACCOUNTED

Confidence decays while you predict and recovers when you measure.

Prediction is where the physics lives: the motion model carries the state forward, and the model's own error inflates the variance.

Each measurement pulls that variance back down. The filter is therefore not just an estimate — it is an estimate **plus a running account of how much to trust it**.

08 · WHERE YOU WILL MEET IT

Same two steps. Four different pairs of sensors.

FAILURE MODES CHOSEN NOT TO OVERLAP

01 MOBILE ROBOT	**IMU + WHEEL ODOMETRY**	Odometry is smooth but drifts with every slip; the IMU is fast but noisy. Fused, heading stays honest between landmarks.
02 DRONE	**GPS + INERTIAL**	Inertial integration is smooth and drifts without bound; GPS is absolute and jumps. One corrects exactly what the other gets wrong.
03 WEARABLE	**9-AXIS IMU + EMG RIG**	Gyroscope, accelerometer and magnetometer each fail differently, so joint angle comes from the fused estimate — never from one axis.
04 TRACKING	**CAMERA DETECTION + MOTION MODEL**	The detector drops frames and occasionally hallucinates. The filter fills the gaps and refuses the physically implausible jump.

A filter is not a trick for merging sensors. It is a rule for how much evidence it takes to change your mind.

SENSOR FUSION · CS 460/560 · NEXT: TUNING Σ^2_{PRED} ON REAL HARDWARE